

End-to-End Machine Learning Solution for Customer Churn Prediction

Abstract

This project presents a complete machine learning pipeline to predict customer churn. It includes data collection, preprocessing, model training, evaluation, and deployment using a web interface.

Introduction

Customer churn is a major problem in telecom industries. Predicting churn helps businesses take proactive measures.

System Architecture

The system consists of: 1. Data Collection 2. Data Preprocessing 3. Model Training 4. Evaluation 5. Deployment

Data Preprocessing

Categorical variables are encoded using Label Encoding. Missing values are removed. Features and target are separated.

Model Training

A Random Forest Classifier is used for prediction. The dataset is split into training and testing sets.

Evaluation

The model is evaluated using accuracy, precision, recall, and F1-score.

Deployment

A Flask web application is built where users can input customer data and get predictions.

Conclusion

This system demonstrates a complete ML lifecycle and can be extended with better models and deployment strategies.